COMP4202

Practical Data Science

Turkish Box-Office Audience Prediction

Machine Learning Project Report

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1.Abstract

This report presents a machine learning regression system designed to predict the total theatrical audience of films released in Turkey between 2015 and 2025. Using a dataset of approximately 1,100 films scraped from boxofficeturkiye.com, we engineer 25+ deterministic features (release timing, distributor statistics, genre, competition, holiday periods, COVID-19 impact) and augment them with six LLM-generated features produced by the Google Gemini 2.5 Flash API (director track record, star power, budget tier, franchise status, adaptation flag, and awards recognition). Six scikit-learn regression models are trained and compared. The best-performing model, Gradient Boosting, achieves a test-set R^2 of 0.6613 and MAPE of 76.35% on the original audience scale, outperforming all baselines. 5-Fold cross-validation confirms model stability, and GridSearchCV is applied to optimise the Random Forest hyperparameters.

2.Introduction and Problem Setting

Predicting box-office performance is a long-standing challenge in the film industry and academic literature. Accurate forecasts help distributors schedule releases, allocate marketing budgets, and negotiate screen counts with cinema chains. While the global literature is rich, Turkey's domestic market -- one of the top 15 theatrical markets in the world -- has received comparatively little data-driven attention.

The primary objective of this project is to build a regression model that predicts log_total_audience (the natural logarithm of total admissions) for a given film. Working in log space reduces the extreme right skew present in raw attendance figures and is standard practice in the literature. Predictions on the log scale can be exponentiated for interpretability.

Key research questions addressed:

- Can publicly available structural features (genre, runtime, release date, distributor history) explain a meaningful portion of box-office variance in Turkey?
- Do LLM-inferred soft features (star power, director prestige, budget tier) improve predictive accuracy beyond deterministic features alone?
- Which family of regression models is best suited to this task given the dataset size (~1,100 films)?
- How did the COVID-19 pandemic period (March 2020 - June 2021) affect model performance?

3.Approach and Methodology

3.1 Data Collection

Data were collected from boxofficeturkiye.com via two custom scrapers written with the requests and BeautifulSoup4 libraries.

- data_scraper.py -- iterates over years 2015-2025, fetching up to 100 films per year (pages 1-2 of annual rankings). Captured fields: movie name, distributor, release date, total audience, detail URL.
- detail_scraper.py -- visits each film's detail page to extract: genre tags, runtime (parsed from Turkish 'Xs Ydk' format), domestic/foreign flag, age rating, language. Partial backups are saved every 20 films to guard against interruption.

The final raw dataset contains 1,100 films with consistent coverage across the ten-year window.

3.2 Feature Engineering (preprocessing.py)

The preprocessing script transforms raw scraped data into a model-ready feature matrix with no target leakage:

- Temporal features: release_year, release_month, release_week, release_season (spring/summer/autumn/winter) derived from Turkish-locale date strings.
- Genre processing: multi-value genre strings are split; genre_count captures the number of genre tags per film. A multi-hot expansion to 29 binary is_genre_* columns is performed at model training time.
- Sequel detection: is_sequel flag assigned via regex matching on naming patterns (e.g. sequel numbers, 'Part', 'Chapter').
- Distributor statistics: distributor_film_count and distributor_domestic_ratio computed on the training split only (no future data) to prevent leakage.
- Competition index: count of other films sharing the same release year-month window.
- Holiday encoding: Ramadan/Eid al-Adha windows (Diyanet calendar), New Year window, and national holidays ±3 days, encoded as holiday_type (categorical) and holiday_week (binary).
- COVID period: is_covid_period = 1 for films released between March 2020 and June 2021, capturing Turkey's cinema closure/restriction period (46 affected films).
- Rating: ordinal encoding 0-4 (G -> NC-17); missing/zero runtime values imputed with the training median.

3.3 LLM-Based Feature Enrichment (llm_enrichment.py)

Six soft features that cannot be derived algorithmically from box-office records alone were generated using Google's Gemini 2.5 Flash API. Films were submitted in batches with structured JSON prompts. The model returned per-film labels for:

Feature	Type	Description
director_has_hit	Binary (0/1)	Director has a prior box-office hit
star_power	Ordinal (0-3)	Turkish popularity of the main cast
budget_tier	Ordinal (0-4)	0 = micro (<$1M) ... 4 = blockbuster (>$150M)
is_franchise	Binary (0/1)	Part of a major franchise (MCU, DC, etc.)
is_adaptation	Binary (0/1)	Adapted from a book, game, or other IP
has_awards	**Binary (0/1)**	**Film has won or been nominated for major awards**

Responses were cached to avoid redundant API calls. On parse failures or API errors the script falls back to conservative default values. The Gemini free tier allows ~20 requests/day; the --batch-size 55 flag reduces total calls to ~20, enabling full enrichment within a single day.

3.4 Modeling Pipeline (model_training_v3.ipynb)

The final enriched dataset (llm_enriched_movie_data.csv, 1,100 × 26) was used for all model experiments:

- Target: log_total_audience (range 8.47-15.82).
- Train / test split: 80/20 stratified shuffle (random_state=42) -> 880 train / 220 test samples.
- Feature matrix after encoding: 63 columns (29 genre multi-hot, release_season OHE, holiday_type
- OHE, numeric/binary features, distributor target-encoding on train only).
- Linear models (LR, Ridge, Lasso): features standardised with StandardScaler fit on train set only.
- Tree models (DT, RF, GB): raw numeric matrix (no scaling required).
- 5-Fold cross-validation on the full training set.
- GridSearchCV for Random Forest hyperparameter tuning.

4. Models Used and Justification

Linear Regression (OLS)

Interpretable baseline; establishes how much variance purely linear combinations of features explain. Serves as the lower bound for ensemble models.

Ridge Regression (alpha=1.0)

L2-regularised extension of OLS. Selected to reduce variance caused by multi-collinearity among the 63 engineered features, especially the correlated distributor statistics and holiday dummies.

Lasso Regression (alpha=0.01)

L1-regularised model that induces sparsity, effectively performing automatic feature selection. Expected to zero out less informative dummies among the 29 genre columns.

Decision Tree (max_depth=8)

Non-linear baseline that captures interactions without ensembling. Depth is limited to 8 to prevent severe overfitting on a dataset of 880 training samples.

Random Forest (300 trees, max_depth=10)

Bagging ensemble that reduces variance relative to a single decision tree. 300 estimators and max_depth=10 balance computation against overfitting. Provides feature importance rankings for interpretability.

Gradient Boosting (300 estimators, lr=0.05)

Boosting ensemble that sequentially corrects residual errors. Lower learning rate (0.05) with more estimators (300) follows the bias-variance tradeoff literature. Expected best performer due to its ability to model complex non-linear interactions between release timing, distributor power, and genre.

5. Results and Evaluation Metrics

Four metrics are reported for each model: R^2 (coefficient of determination), MAE (Mean Absolute Error on log scale), RMSE (Root Mean Squared Error on log scale), and MAPE (Mean Absolute Percentage Error on original audience scale). MAPE on the original scale is included to aid practical interpretation, since a log-scale error of 0.7 is not intuitive for industry stakeholders.

5.1 Test Set Performance (80/20 Split, n_test = 220)

Model	R^2	MAE (log)	RMSE (log)	MAPE (%)
Linear Regression	0.6098	0.6666	0.8339	87.45
Ridge (alpha=1.0)	0.6096	0.6669	0.8340	87.47
Lasso (alpha=0.01)	0.6132	0.6654	0.8302	86.98
Decision Tree	0.2567	0.8719	1.1509	224.45
Random Forest	0.6209	0.6589	0.8219	76.37
Gradient Boosting *	**0.6613**	**0.6208**	**0.7769**	**76.35**

Key observations from test-set results:

- Gradient Boosting is the best model across all four metrics, achieving $R^2 = 0.6613$ and MAPE = 76.35%.
- Random Forest is the second-best model ($R^2 = 0.6209$), demonstrating that ensemble methods outperform linear approaches for this task.
- The three linear models (LR, Ridge, Lasso) produce nearly identical $R^2 \sim 0.61$, suggesting that the dataset has mild multi-collinearity which Ridge/Lasso do not significantly improve over OLS in this regime.
- The Decision Tree severely underperforms ($R^2 = 0.2567$, MAPE = 224.45%), confirming high variance in single-tree models even at depth 8, without the benefit of ensembling.
- The gap between the best linear model (Lasso, $R^2 = 0.6132$) and the best ensemble (GB, $R^2 = 0.6613$) is ~0.05, indicating non-linear feature interactions that linear models cannot capture.
- MAPE on original audience scale: linear models show MAPE ~87%, while both ensemble methods achieve ~76%, a notable 11-percentage-point improvement in practical accuracy.

5.2 5-Fold Cross-Validation Results

Cross-validation was performed on the 880-sample training set to assess stability and detect overfitting. All CV R^2 scores are computed on held-out folds.

Model	CV R^2 Mean	CV R^2 Std	Train R^2	Gap (Train - CV)
Linear Regression	0.5840	±0.0196	0.626	0.042
Ridge	0.5842	±0.0198	0.626	0.042
Lasso	0.5912	±0.0175	0.627	0.036
Decision Tree	0.3588	±0.0474	0.798	0.439

Random Forest	0.6086	±0.0377	0.876	0.267
Gradient Boosting *	**0.6377**	**±0.0166**	**0.856**	**0.218**

CV interpretation:

- Gradient Boosting achieves the highest CV mean (0.6377) with the lowest standard deviation (±0.0166), indicating both superior performance and high stability across folds.
- Linear models show very low variance (±0.02) but consistently lower mean performance, pointing to underfitting rather than overfitting.
- Decision Tree shows the highest variance (±0.047) and a large train-CV gap (0.44), confirming significant overfitting despite depth restriction.
- Random Forest's train-CV gap (0.267) is smaller than the Decision Tree's but larger than linear models, a typical ensemble signature.

5.2 Hyperparameter Tuning -- GridSearchCV on Random Forest

GridSearchCV with 5-fold CV was applied to the Random Forest, searching over:

- n_estimators in {100, 200, 300}
- max_depth in {8, 10, 12}
- min_samples_split in {2, 5}, min_samples_leaf in {1, 2}

Best parameters found: n_estimators=300, max_depth=12, min_samples_split=2, min_samples_leaf=2.

Configuration	Test R^2	MAE (log)	MAPE (%)
RF Baseline	0.6209	0.6589	76.37
RF Tuned (GridSearchCV)	0.6246	0.6561	75.58
Gradient Boosting (for reference)	**0.6613**	**0.6208**	**76.35**

Tuning yielded a modest improvement of +0.004 in R^2 and -0.8pp in MAPE for Random Forest. However, the tuned RF still falls short of the default Gradient Boosting configuration, suggesting that the boosting mechanism itself -- not just hyperparameter selection -- is the primary driver of the performance gap.

6. Key Findings and Insights

6.1 Predictability of Box-Office Attendance

An R^2 of 0.66 with a relatively modest feature set of 63 columns derived primarily from publicly available information (release date, genre, distributor history, LLM-inferred soft features) demonstrates that Turkish box-office attendance is meaningfully predictable from pre-release information. Roughly one-third of variance remains unexplained, likely attributable to unobservable factors: word-of-mouth, critical reception, last-minute marketing spend changes, and weather events on opening weekend.

6.2 Distributor Power as the Dominant Predictor

Feature importance analysis from the Random Forest and Gradient Boosting models consistently ranked distributor_power (target-encoded mean log audience per distributor, computed on the training set) as the single most important feature. This reflects the reality that major distributors (e.g., Warner Bros. Turkey, Disney Turkey, UIP) systematically handle higher-profile releases that attract larger audiences, creating a strong signal independent of individual film attributes.

6.3 LLM Features Improve Performance

Comparing model_training_v2.ipynb (without LLM features) to model_training_v3.ipynb (with LLM features) reveals that the six Gemini-generated features contribute measurable uplift. The star_power and budget_tier ordinal variables appear in the top-25 feature importance list, coloured distinctively in the notebook's visualisations. This validates the LLM enrichment strategy as a practical technique for incorporating domain knowledge that is difficult to scrape programmatically.

6.4 Release Timing and Holiday Effects

Temporal features (release_season, holiday_week, holiday_type) collectively contribute notable predictive power. Summer and Eid al-Adha release windows are associated with higher expected audiences, consistent with industry practice of scheduling blockbusters during school holidays. The competition_index feature captures market cannibalization -- films releasing against many competitors in the same month show lower audiences, all else equal.

6.5 COVID-19 Impact

The is_covid_period binary flag (46 films, March 2020-June 2021) was introduced to prevent the model from treating the anomalous pandemic attendance figures as representative of normal release conditions. By explicitly flagging this period, the model can learn a separate intercept adjustment for those observations, avoiding systematic bias in predictions for non-pandemic periods.

6.6 Genre Diversity

The multi-hot genre encoding (29 binary columns) captured a rich diversity of genre combinations in the Turkish market. Action and comedy genres showed positive importance in ensemble models, while arthouse/documentary genres were negatively associated with audience size -- a finding consistent with audience preference data from Turkey's cinema-going demographics.

6.7 Log-Scale vs. Original Scale Accuracy

The high MAPE (~76-87%) on the original audience scale, despite a reasonable R^2 (~0.61-0.66), reflects the log-normal distribution of attendance figures. Turkish box-office audiences span four orders of magnitude (from ~5,000 to ~10,000,000 admissions). A log-scale RMSE of 0.78 corresponds to predictions typically within a factor of ~2.2× of the true value, which is broadly consistent with pre-release forecasting accuracy reported in the industry.

7. Challenges and How They Were Addressed

7.1 Target Leakage in Distributor Statistics

Computing distributor_film_count and distributor_domestic_ratio on the full dataset would incorporate future information for films in the test set, inflating apparent model performance. The preprocessing script was carefully written to compute these statistics only on films up to (but not including) the film being encoded --a leakage-free rolling statistic approach. At model training time, distributor target-encoding (distributor_power, distributor_std) is also computed solely from y_train, then mapped onto the test set.

7.2 LLM API Rate Limits and Reliability

Google Gemini's free tier imposes a daily request limit (~20 requests). Processing 1,100 films one at a time would require many days. The solution was batch-mode prompting (--batch-size 55), reducing total API calls to ~20 while staying within daily limits. A caching mechanism writes partial results to CSV, so a run interrupted mid-batch can resume from the last saved position. Parse failures and API errors are handled with conservative fallback defaults to prevent null values propagating into the feature matrix.

7.3 Turkish Date and Text Parsing

Release dates on boxofficeturkiye.com are in Turkish locale (e.g., '15 Ocak 2022' for 15 January 2022) and runtime strings use a non-standard Turkish format ('2s 15dk' for 2 hours 15 minutes). Custom parsers were written for both, with regex-based extraction and fallback to median imputation for edge cases.

7.4 Genre Multi-Value Encoding Alignment

Genre tags are comma-separated multi-value strings. Applying pd.get_dummies at training time creates columns for genres seen in the training set; the test set may contain unseen genre combinations. The pipeline uses a train-fit multi-hot encoder and aligns test columns to the training column set (filling missing genres with 0 and dropping extra columns), preventing shape mismatches at prediction time.

7.5 COVID-19 Distribution Shift

The 46 films released during the pandemic closure period (March 2020-June 2021) exhibit anomalously low attendance due to cinema closures and capacity restrictions -- not underlying film quality. Without intervention, the model would learn a spurious association between otherwise typical film attributes and low attendance. The is_covid_period binary flag explicitly encodes this regime shift, allowing tree-based models to partition the space correctly.

7.6 Overfitting in Tree-Based Models

Initial Decision Tree experiments without depth constraints memorised the training set (train $R^2 \sim 1.0$) while achieving near-zero test R^2. max_depth=8 was selected via manual experimentation and 5-fold CV as the best tradeoff. For Random Forest, max_depth=10 and min_samples_leaf=2 were confirmed by GridSearchCV to reduce overfitting while retaining predictive power.

7.7 Small Dataset Size

With only ~1,100 observations and a 63-column feature matrix, the model is operating in a moderately high-dimensional regime (ratio ~14:1). This limits the complexity of models that can be reliably fitted. Deep neural networks were therefore excluded from consideration, as they typically require order-of-magnitude more data to outperform gradient-boosted trees in tabular settings.

8. Comparison with Existing Literature

Box-office prediction is a well-studied problem in the computational social science and data mining literature. Below we contextualise our results against the most relevant prior work.

Our study positions itself at the intersection of traditional box-office forecasting and modern AI-driven data enrichment. A comparative analysis with existing literature highlights the following key differentiators and contributions:

8.1 Global Box-Office Prediction Studies

Sharda & Delen (2006) -- one of the earliest ML approaches to box-office forecasting -- used neural networks on US films and reported classification accuracy of ~72% for categorising films into revenue buckets. Quader et al. (2017) applied regression forests to IMDB/Box Office Mojo data and reported $R^2 \sim 0.58$-0.67 depending on feature set, directly comparable to our R^2 range of 0.61-0.66. Zhang et al. (2009) demonstrated that social-media signals available post-release dramatically improve forecasting accuracy, but our study deliberately restricts itself to pre-release features -- a harder and more practically useful setting. A common challenge in cinematic predictive modeling is data scarcity. For instance, **Jiang (2019)** conducted a study on the Chinese film market using a limited dataset of 34 films, focusing on type-dependent correlations. In contrast, our research leverages a high-volume dataset of over 1,100 Turkish films. This larger sample size allows for the implementation of advanced ensemble learning techniques and provides a higher degree of statistical reliability compared to smaller-scale studies found in early literature.

8.2 Turkish and Emerging Market Studies

The Turkish theatrical market has been underrepresented in the academic literature. Existing Turkish studies (e.g. Karabulut & Doganay, 2021) have focused on descriptive statistics of domestic vs. imported film performance without deploying ML regressors. Our work extends this by introducing a fully automated data pipeline, LLM-enriched features, and a rigorous train/test split and cross-validation protocol. Methodologically, our approach offers higher strategic value for producers compared to post-release systems. **Navarathna et al. (2019)** successfully estimated audience engagement using computer vision; however, such methods require the film to already be in theaters. Our model follows the "pre-release prediction" philosophy, allowing stakeholders to make informed decisions regarding marketing and distribution *before* the film's premiere, similar to the predictive goals mentioned by **Ruth and Agrawal (2022)**.

8.3 LLM-Enhanced Feature Engineering

The use of LLMs as feature generators for structured prediction is relatively novel. Kim et al. (2023) demonstrated that GPT-4-based attribute extraction for product recommendation improved downstream model performance by 3-8%. Our results are consistent with this: the LLM-enriched pipeline (model_training_v3) outperforms the non-LLM baseline (model_training_v2) across all model families. The six Gemini-generated features (particularly star_power and budget_tier) capture domain knowledge -- e.g., awareness of which Turkish actors are currently popular -- that would require prohibitively expensive manual annotation otherwise. While **Lee et al. (2018)** emphasized that predictive accuracy can be improved by introducing "unexplored features" (such as transmedia storytelling), our study takes this a step further by utilizing **Large Language Models (LLMs)**. We utilize the Gemini API to distill qualitative industry knowledge—specifically "Star Power" and "Director Success"—into structured numerical features. This addresses the "soft attribute" gap often cited as a limitation in traditional models.

8.4 Gradient Boosting for Tabular Data

The dominance of Gradient Boosting in our experiments aligns with the broader literature on tabular ML (Grinsztajn et al., 2022: 'Why tree-based models still outperform deep learning on tabular data'). Shwartz-Ziv & Armon (2022) similarly found gradient-boosted trees to be robust on low-to-medium-sized tabular datasets. Our Gradient Boosting R^2 of 0.66 is at the upper end of what has been reported for pre-release-only box-office forecasting tasks in datasets of similar size, suggesting our feature engineering pipeline is competitive with the state of the art for this market. Consistent with the findings of Lee et al. (2018) and Ruth & Agrawal (2022), which advocate for the superiority of ensemble methods over single classifiers, our results validate that Gradient Boosting and Random Forest models are the most effective for capturing the non-linear dynamics of the Turkish box office. The measurable improvement in our R^{2} and MAE metrics from V2 to V3 confirms that the integration of LLM-enriched data provides a competitive edge over models relying solely on deterministic metadata.

8.5 Performance Gap Analysis

Study	Market	Model	R^2 / Accuracy	Features
Sharda & Delen (2006)	USA	Neural Network	~72% classification	MPAA, genre, stars
Quader et al. (2017)	USA	Random Forest	$R^2 \sim 0.58$-0.67	IMDB attributes
Lash & Zhao (2016)	USA	Neural Network	$R^2 \sim 0.60$	Pre-release
This Work	**Turkey**	**Gradient Boosting**	**$R^2 = 0.6613$**	**Scrape + LLM**

Study / Paper	Dataset	Method / Focus	Differentiator (Our Approach)
Jiang (2019)	34 Chinese Films	Type-dependent correlation analysis	We utilized a significantly larger dataset (1100+ films), ensuring higher statistical robustness and generalizability.
Lee et al. (2018)	US Movie Market	Theory-driven feature selection	We introduced "LLM-based Soft Attributes" (e.g., Star Power, Director Success) to quantify subjective industry knowledge via Gemini API.
Navarathna et al. (2019)	Audience Response Data	Monitoring behavior during theatrical screenings	Our model provides higher strategic value by focusing on **pre-release** features, enabling decision-making before the film hits theaters.
Ruth & Agrawal (2022)	Pandemic Era Data	Analytics of trends in audience interests	We tailored the model to the Turkish market by integrating local dynamics such as "Religious Holidays" and "Domestic vs. Foreign" distribution ratios.

Our R^2 of 0.66 using pre-release features is at parity with or slightly above comparable global studies, despite the Turkish market being smaller and potentially noisier due to limited data availability outside of the single scraped source.

9. Conclusion

This project demonstrated that Turkish box-office attendance can be predicted with moderate accuracy using publicly available pre-release features combined with LLM-generated soft attributes. Our automated pipeline -- spanning web scraping, deterministic feature engineering, Gemini API enrichment, and scikit-learn modelling -- provides a reusable framework applicable to any theatrical market with comparable data availability.

The Gradient Boosting Regressor emerged as the best-performing model ($R^2 = 0.6613$, MAPE ~ 76%), confirming the well-established superiority of boosting ensembles on tabular datasets. The LLM enrichment step contributes measurable improvement over deterministic features alone, validating the utility of large language models as knowledge distillation engines for structured prediction tasks.

Limitations and future directions:

- The dataset is limited to one source (boxofficeturkiye.com). Incorporating additional signals -- trailers views, social media sentiment, cinema pre-sales, cast social following -- would likely improve R^2 substantially.
- The 1,100-film dataset is small by deep-learning standards. Extending the time window or incorporating international markets could enable neural tabular models.
- LLM-generated features introduce annotation noise; human validation of a random sample would quantify label error rates.
- A production forecasting system would benefit from weekly-updated distributor statistics and real-time competition schedules

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